

Five approaches to smooth limits of prime-degree plane curves

Partial results and exact remaining gaps for problem 30005171

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Abstract

The assertion that every smooth abstract limit of prime-degree plane curves embeds as a Cartier divisor in a log-terminal $\mathbb{Q}$ -Gorenstein degeneration of $\mathbb{P}^2$ is not proved or disproved here. We give five substantive approaches, with complete arguments for the partial statements obtained: a limiting-linear-series obstruction; a hyperelliptic test family showing the weakness of root and Hilbert-function tests; a Cartier-index and genus restriction; a nodal-boundary exclusion; and an exhaustive degree-11 rational-unicuspidal threshold obstruction. The last argument has a short mathematical exhaustion proof independent of its computational replay. All general geometric dependencies are stated explicitly. No novelty or priority claim is made.

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1 Question, status, and scope

Throughout, the ground field is $\mathbb{C}$. A smooth limit means the central fiber C_0 of a smooth proper family of connected curves over a smooth pointed curve, with general fiber a smooth plane curve of degree p . After shrinking and finite base change we work over a discrete valuation ring. Set

$$g_p = \frac{(p-1)(p-2)}{2}.$$

A $\mathbb{Q}$ -Gorenstein degeneration of $\mathbb{P}^2$ here is an actual flat surface family with general fiber $\mathbb{P}^2$ and relatively $\mathbb{Q}$ -Cartier canonical divisor, with the usual compatible reflexive powers. Merely writing down a quotient surface with the same canonical self-intersection is insufficient.

The source is Conjecture B in DeVleming's contribution to Oberwolfach Report 32/2022, printed p. 1822 [1]. It asks for a Cartier embedding of every such C_0 in a log-terminal $\mathbb{Q}$ -Gorenstein degeneration of $\mathbb{P}^2$. The published paper [2], p. 685, states the same conjecture using Manetti surfaces. We use the following established results, not as new contributions:

- Hacking–Prokhorov's classification identifies the relevant surfaces with partial smoothings of $\mathbb{P}(a^2, b^2, c^2)$ for positive Markov triples $a^2 + b^2 + c^2 = 3abc$ [3, Cor. 1.2].
- The prime cases 2, 3 are elementary; [2, Thms. D,E] establishes the cases 5, 7.
- Nonplanar limits constructed in Markov degrees in [2, Thm. 1.10] are Cartier curves on these surfaces. They do not contradict the conjecture.

Targeted searches and the authors' current research lists, including the 2026 preprints [6, 7], did not yield a general resolution. This is a bounded literature search, not a certificate that no result exists elsewhere. The degree-11 cusp computation below is consistent with the prior classification [5, Thm. 5.1]; it is an independently written verification, not a claim to discover a new classification.

For clarity, four objects are distinct: the smooth abstract curve C_0 ; its possibly degenerate limiting plane linear series; the possibly singular divisor D_0^{CY} in a Calabi–Yau limit of plane pairs; and the singular ambient surface X_0^{CY} . The existence of the last two with $\text{lct}(X_0^{\text{CY}}, D_0^{\text{CY}}) \geq 3/p$ does not identify D_0^{CY} with C_0 . Likewise, a smooth Weil divisor on a singular surface need not be Cartier.

2 Approach 1: extend the plane polarization

The attempted proof is to extend the degree- p plane linear series across the smooth family and show that it still embeds. The following gives precisely what this method proves.

Lemma 2.1 (Existence of a limiting net). *Let $f : \mathcal{C} \rightarrow \text{Spec } R$ be a smooth proper family of connected curves over a DVR, with general fiber a smooth degree- p plane curve, $p \geq 5$. After a finite base extension needed to define the generic plane embedding, there is a line bundle L on $\mathcal{C}$ such that*

$$\deg L_0 = p, \quad L_0^{\otimes(p-3)} \simeq \omega_{C_0}, \quad V \subseteq H^0(C_0, L_0), \quad \dim V = 3.$$

The net V is the specialization of the three generic plane-coordinate sections. It may have base points.

Proof. The total space is a regular surface. Represent $\mathcal{O}_{C_\eta}(1)$ by a divisor on the generic curve and take its closure; a divisor on a regular surface is Cartier. This produces L , of constant fiber degree p . Generic plane adjunction gives $L_\eta^{p-3} \simeq \omega_{C_\eta}$. A generic trivialization of $L^{p-3} \otimes \omega_{C/R}^{-1}$ has divisor supported on the special fiber. That fiber is irreducible and is the principal divisor of a uniformizer, so its associated line bundle restricts trivially. This proves the asserted root identity.

The finite R -module f_*L is torsion-free and hence free, of rank $h^0(C_\eta, \mathcal{O}(1)) = 3$. The cohomology sequence for multiplication by a uniformizer gives an injection

$$(f_*L) \otimes_R R/\mathfrak{m} \hookrightarrow H^0(C_0, L_0).$$

Its image is the required three-dimensional net. Notice that $h^0(C_0, L_0)$ itself may be larger than three. $\square$

Proposition 2.2 (Prime degree and the fixed part). *Let C be smooth of genus g_p , where $p \geq 5$ is prime. If a degree- p line bundle L has a base-point-free three-dimensional linear series, then that series embeds C as a smooth degree- p plane curve. If C is nonplanar and (L, V) is as in Lemma 2.1, let B be the common zero divisor of V , $b = \deg B$, and write m for the degree of the induced map to its integral plane image of degree e . Then*

$$1 \leq b \leq p - 4, \quad m \geq 2, \quad e \geq 2, \quad p - b = me.$$

Proof. Removing B gives a base-point-free net in $L(-B)$, hence a nonconstant morphism $C \rightarrow \mathbb{P}^2$. The image is integral and nondegenerate: a linear equation of the image would be a linear dependence among the three sections. Thus $e \geq 2$. Pulling back a general line gives $p - b = me$.

If $b = 0$, primality implies $m = 1$ and $e = p$. The map from the smooth proper curve is finite birational and hence is its image's normalization. Plane adjunction and the normalization exact sequence give

$$g(C) = g_p = p_a(\text{image}) = g(C) + \sum_Q \delta_Q.$$

All δ_Q vanish. The integral plane image is normal, hence smooth, and the normalization map is an isomorphism.

If $b > 0$ and $m = 1$, the same argument gives $g_p = g(C) \leq (e - 1)(e - 2)/2 < g_p$, since $e \leq p - b < p$. This is impossible. Therefore $m, e \geq 2$, so $p - b \geq 4$, as claimed. $\square$

Corollary 2.3. *A nonplanar smooth limit of plane quintics is hyperelliptic.*

Proof. For $p = 5$, the proposition forces $b = 1$ and $me = 4$, so $m = e = 2$. The image is an integral conic and hence is $\mathbb{P}^1$. The degree-two map is a hyperelliptic pencil. $\square$

This is an abstract-curve argument, and does not by itself put that curve on a Manetti surface. The Cartier realization required by the original question is supplied in degree five by the cited work.

For $p = 11$, the possible triples (b, m, e) from this method are

$$(1, 2, 5), (1, 5, 2), (2, 3, 3), (3, 2, 4), (3, 4, 2), \\ (5, 2, 3), (5, 3, 2), (7, 2, 2).$$

These are necessary numerical possibilities, not existing smoothings. For example, Riemann–Hurwitz for a double cover of a genus- $h \leq 6$ normalization of a plane quintic gives ramification degree $92 - 4h$, which is nonnegative. Therefore that elementary test does not eliminate the first case.

Exact remaining gap. There is no proof that the limiting net is base-point-free. Removing its base divisor changes its degree and destroys the prime-factor argument. Moreover a proof of base-point-freeness for every prime would be false because the known quintic and other Markov-prime examples are nonplanar. One needs to interpret the base locus in an ambient surface degeneration, rather than discard it.

3 Approach 2: canonical roots and Hilbert functions

The second attempted route is to use the root identity in Lemma 2.1, together with every semi-continuity inequality for powers of the polarization. A family of explicit test objects demonstrates that these conditions do not suffice.

Proposition 3.1 (Hyperelliptic numerical test objects). *For every odd integer $p \geq 5$, let C be any smooth hyperelliptic curve of genus g_p , and let P be a Weierstrass point. Set $L = \mathcal{O}_C(pP)$. Then*

$$L^{p-3} \simeq \omega_C, \quad \deg L = p,$$

and, for every integer $k \geq 0$,

$$h^0(C, L^k) \geq h^0(C_p^{\text{pl}}, \mathcal{O}_{C_p^{\text{pl}}}(k)),$$

where C_p^{pl} is a smooth plane curve of degree p . The complete series $|L|$ has fixed divisor exactly P , and its moving part factors through the hyperelliptic map.

Proof. Write A for the degree-two hyperelliptic pencil. Since P is a ramification point, $A \simeq \mathcal{O}_C(2P)$, and hyperelliptic adjunction gives

$$\omega_C \simeq A^{g_p-1} = \mathcal{O}_C((2g_p - 2)P), \quad 2g_p - 2 = p(p - 3).$$

This is the root identity.

Choose an affine equation $y^2 = f(x)$ with $\deg f = 2g_p + 1$ and P the unique point at infinity. The pole orders of x, y are $2, 2g_p + 1$. The ring of functions regular off P is $\mathbb{C}[x, y]/(y^2 - f(x))$, with vector-space basis the monomials x^i and $x^i y$. In particular, for $0 \leq n \leq 2g_p$,

$$h^0(C, \mathcal{O}_C(nP)) = \lfloor n/2 \rfloor + 1.$$

For $0 \leq k \leq p - 3$, we have $kp \leq 2g_p - 2$, while the plane restriction sequence gives $h^0(C_p^{\text{pl}}, \mathcal{O}(k)) = (k + 1)(k + 2)/2$. The difference is

$$\left\lfloor \frac{kp}{2} \right\rfloor + 1 - \frac{(k + 1)(k + 2)}{2} = \left\lfloor \frac{k(p - k - 3)}{2} \right\rfloor \geq 0.$$

For $k \geq p - 2$, both line bundles are nonspecial, because $kp > 2g_p - 2$. Their dimensions equal $kp - g_p + 1$ by Riemann–Roch. This proves all the inequalities, including arbitrarily large k .

Because $p < 2g_p + 1$ and p is odd, a basis for $H^0(C, L)$ is $1, x, \dots, x^{(p-1)/2}$. Their maximum pole order at P is $p - 1$, so they vanish as sections of L at P . The last basis vector has precisely first-order vanishing there; the section 1 has no zero away from P . Thus the common zero divisor is exactly P . The moving series is $|(p - 1)P| = |A^{(p-1)/2}|$, which factors through the hyperelliptic map. $\square$

There is also an explicit net with moving degree four: the subspace spanned by $1, x, x^2$ has fixed divisor $(p - 4)P$ and maps C two-to-one onto a conic. For $p = 7$, these test objects satisfy every condition just proved, yet [2, Thm. E] excludes them as smooth limits of plane septic. This is a direct known negative control for the attempted sufficiency criterion. It does not assert that this chosen net is the specialization of a plane net.

Exact remaining gap. The graded spaces have the required dimensions, but their multiplication maps and their deformations as a polarized curve are not controlled. We have neither a flat deformation of the graded algebra to a plane-curve algebra nor a smoothing of any of these test objects to a plane curve for a new prime. The construction is not a counterexample to the question.

4 Approach 3: Cartier indices, genus, and ambient surfaces

We now assume an ambient surface exists, and test what it can be. For a Manetti surface X , use the following class-group facts from [2, Thm. 1.6]: there is an ample positive generator H of $\text{Cl}(X) \simeq \mathbb{Z}$ and a positive integer α such that

$$H^2 = \frac{1}{\alpha^2}, \quad \text{Pic}(X) = \mathbb{Z} \cdot \alpha^2 H.$$

The integer α is the product of the unsmoothed Markov entries. Since $K_X^2 = 9$ and $-K_X$ is ample, $K_X = -3\alpha H$ in this torsion-free group.

Lemma 4.1 (Smooth Cartier curves avoid singular points). *If a Cartier divisor on a surface is smooth at a closed point, the surface is smooth at that point.*

Proof. Let $(R, \mathfrak{m})$ be the local ring of the surface, and let a nonzerodivisor $f \in \mathfrak{m}$ define the divisor. The ring $R/(f)$ is regular of dimension one. Its maximal ideal has one generator; lifting that generator and adding f generates $\mathfrak{m}$ by Nakayama's lemma. Thus $\dim_{\mathbb{C}} \mathfrak{m}/\mathfrak{m}^2 \leq 2 = \dim R$, and R is regular. Over $\mathbb{C}$ this is smoothness. $\square$

The word Cartier is essential: on $\text{Spec } \mathbb{C}[u, v]^{\mu_r}$ with scalar action $(u, v) \mapsto (\zeta u, \zeta v)$, the image of $u = 0$ is the smooth curve $\text{Spec } \mathbb{C}[v^r]$, while its nontrivial local divisor class prevents it from being Cartier for $r > 1$.

Proposition 4.2 (Genus forces the prime index). *Let $p \geq 5$ be prime and let $C \subset X$ be a smooth connected effective Cartier divisor of genus g_p on a Manetti surface. Then α divides p . Consequently either $X \simeq \mathbb{P}^2$ or X has exactly one remaining singularity, with local class group of order p^2 ; in the latter case p is a Markov number.*

Proof. Write $C = m\alpha^2 H$, with $m > 0$ since C is nonzero effective and H is ample. Put $q = m\alpha$. By Lemma 4.1, C avoids $\text{Sing } X$, so ordinary adjunction along its smooth neighborhood gives

$$2g_p - 2 = C(C + K_X) = m^2\alpha^2 - 3m\alpha = q(q - 3).$$

Thus

$$(q - p)(q + p - 3) = 0.$$

As $q > 0$ and $p \geq 5$, the second factor is positive; hence $q = p$ and $\alpha \mid p$. If $\alpha = 1$, all singularities have been smoothed and the classified surface is $\mathbb{P}^2$. If $\alpha = p$, the product of the remaining nontrivial Markov entries is the prime p , so there is exactly one such entry, equal to p . $\square$

This proposition only assumes the abstract genus and a Cartier embedding. It does not quietly assume that this particular embedding already extends the given family. Once a compatible degree- p pair family is given, its specialization has class $p\alpha H$, and its being Cartier is equivalently $\alpha^2 \mid p\alpha$, the same divisibility condition.

Lemma 4.3. *The integer 11 is not a Markov number.*

Proof. We recall the elementary descent. Sort a positive Markov triple as $a \leq b \leq c$. Apart from the root $(1, 1, 1)$, replacing c by $c' = 3ab - c = (a^2 + b^2)/c$ gives a positive integer at most b and decreases the maximal entry when $c > b$. Indeed the polynomial $F(t) = t^2 - 3abt + a^2 + b^2$ has roots c, c' and $F(b) = a^2 + (2 - 3a)b^2 \leq 0$; the equality exceptions are checked directly. If $b = c$, then $a^2 = (3a - 2)b^2$ and $a \leq b$ force $a = b = 1$, so the only equal-maximal triple is the root.

If a triple contains 11, mutate only maximal entries greater than 11 until its maximum is 11. It remains to solve $a^2 + b^2 + 121 = 33ab$ with $1 \leq a \leq b \leq 11$. As $F(b) \leq 0$ in the preceding argument, descent of this bounded set gives precisely $(1, 1, 1)$, $(1, 1, 2)$, $(1, 2, 5)$ as the triples with maximum at most 11; equivalently, direct checking of the 66 pairs (a, b) in the displayed equation gives no zero. A dependency-free exact check is included. Thus no triple contains 11. $\square$

The finite check can also be read as a compact integer certificate: for each $a = 1, \dots, 11$, the quadratic $b^2 - 33ab + (a^2 + 121)$ has no integer root in $[a, 11]$.

Exact remaining gap. The divisibility condition classifies possible ambient surfaces only after a Cartier embedding is supplied. It does not create the embedding, prove that a Calabi–Yau divisor is Cartier, or show that the smooth abstract limit equals that divisor. For $p = 11$ it implies that a positive answer must force planarity, but it does not establish that planarity.

5 Approach 4: exclude nodal boundary curves

The fourth approach asks whether semistable reduction and positivity alone force a Calabi–Yau divisor to be smooth. They do so in a useful restricted regime.

Theorem 5.1 (Nodal curve in the smooth locus). *Let X be a Manetti surface and let $D \subset X \setminus \text{Sing } X$ be a connected, reduced nodal curve. Suppose it occurs as the special fiber of a proper flat family of curves with smooth general fiber and genus at least two. If the stable model of that family has a smooth special fiber, then D is already smooth and irreducible.*

Proof. The nodal curve has a dual multigraph Γ whose loops include self-nodes. Semistable reduction of a family with nodal central fiber only subdivides node edges if needed; its stable model contracts unstable rational components. Neither operation changes the first Betti number of the graph. The smooth stable fiber has graph with one vertex and no edges, so $b_1(\Gamma) = 0$: the graph of D is a tree.

Write $D = \sum_{i=1}^r D_i$. Each D_i is Cartier, since its support lies in the smooth locus and its ideal is trivial near the surface singularities. Write $D_i = m_i A$ where $A = \alpha^2 H$ is the ample Picard generator and m_i is a positive integer. Distinct components have intersection

$$D_i D_j = m_i m_j A^2 = m_i m_j \alpha^2 > 0.$$

Because D is nodal, this is exactly the number of edges between their vertices. Every pair of distinct vertices is adjacent. A tree with this property has $r \leq 2$.

If $r = 2$, there is exactly one edge, so $m_1 m_2 \alpha^2 = 1$. Hence $\alpha = m_1 = m_2 = 1$, $X = \mathbb{P}^2$, and D is the union of two lines. Its arithmetic genus is zero, contrary to the genus hypothesis. Thus $r = 1$. A self-node would give a loop, again contradicting $b_1 = 0$. The irreducible nodal curve has no nodes and is smooth. $\square$

The graph invariant used here can also be verified by the normalization exact sequence: for a connected nodal curve, $p_a(D) = \sum_i g(\tilde{D}_i) + b_1(\Gamma)$. Subdividing an edge adds one edge and one vertex; contracting a rational tail or suppressing a rational two-valent vertex preserves b_1 . A cycle of rational components cannot disappear into a smooth stable fiber of genus at least two.

Applied to a Calabi–Yau limit under the theorem’s extra hypotheses, uniqueness of stable reduction identifies its smooth special fiber with the originally given C_0 . In this regime, the sought Cartier realization is indeed obtained.

Exact remaining gap. The hypotheses that the divisor is reduced, nodal, and avoids $\text{Sing } X$ are not known for general prime degree. High-order cusps, multiple components, nonnormal total divisor spaces, and curves through quotient singularities are precisely situations where this graph proof does not apply. Log canonicity with coefficient $3/p < 1$ does not imply that a curve is nodal. No such implication is used here.

6 Approach 5: an exhaustive degree-11 cusp obstruction

The fifth approach attacks one non-nodal case left by the preceding theorem, using the log-canonical threshold and the semigroup distribution theorem. It produces a complete exclusion for one particular geometric class.

Theorem 6.1. *If $D \subset \mathbb{P}^2$ is an irreducible rational degree-11 curve whose only singular point is unibranch, then*

$$\text{lct}(\mathbb{P}^2, D) < \frac{3}{11}.$$

Thus such a curve cannot be the divisor of a log-canonical pair $(\mathbb{P}^2, (3/11)D)$.

6.1 Imported singularity formulas

For a plane branch write its characteristic sequence as $(m; \beta_1, \dots, \beta_s)$, with $m \geq 2$, $\beta_1 > m$, and strictly decreasing gcds $e_0 = m$, $e_i = \gcd(m, \beta_1, \dots, \beta_i)$, ending at $e_s = 1$. Set $N_i = \beta_i - \beta_{i-1}$ for $i \geq 2$. The classical formulas used below are

$$2\delta = (m-1)(\beta_1-1) + \sum_{i=2}^s (e_{i-1}-1)N_i, \quad (1)$$

$$\text{lct} = \frac{1}{m} + \frac{1}{\beta_1}. \quad (2)$$

The value semigroup has generators $v_0 = m$, $v_1 = \beta_1$, and

$$v_i = \frac{e_{i-2}}{e_{i-1}}v_{i-1} + N_i \quad (i \geq 2).$$

These formulas are recorded in [5, §2, §3.5]. Only the characteristic exponents are used, not all nonzero terms in a Puiseux expansion. Finally, for a rational unicuspidal degree- d plane curve the Borodzik–Livingston theorem [4, Thm. 6.5] says that

$$R(jd+1) := \#(S \cap [0, jd+1)) = \frac{(j+1)(j+2)}{2}, \quad -1 \leq j \leq d-2. \quad (3)$$

The half-open interval convention is important. These are necessary existence conditions, never sufficient conditions.

6.2 Mathematical exhaustion of the threshold regime

Lemma 6.2. *For a plane branch with $\delta = 45$ and $\text{lct} \geq 3/11$, the complete list of possible characteristic sequences is*

$$\begin{aligned} &(2; 91), (3; 46), (4; 31), \\ &(4; b, 93-2b) \quad (b = 6, 10, 14, 18, 22, 26, 30), \\ &(6; 8, 63), (6; 9, 34). \end{aligned}$$

This list has twelve entries and is exhaustive for every number of Puiseux pairs.

Proof. Since $\beta_1 \geq m + 1$, $m \geq 8$ gives $\text{lct} \leq 1/8 + 1/9 = 17/72 < 3/11$. Thus $2 \leq m \leq 7$. A strictly decreasing gcd chain beginning with one of these integers has at most two drops: the only composite possibilities are $4 \rightarrow 2 \rightarrow 1$, $6 \rightarrow 2 \rightarrow 1$, and $6 \rightarrow 3 \rightarrow 1$. This already rules out three or more characteristic exponents in the entire threshold regime.

For prime $m = 2, 3, 5, 7$, the first gcd drop is directly to 1, so (1) gives $(m - 1)(\beta_1 - 1) = 90$. For $m = 2, 3$ this yields $\beta_1 = 91, 46$, respectively. For $m = 5$ there is no integer solution. For $m = 7$ the solution $\beta_1 = 16$ has threshold $23/112 < 3/11$ and is excluded.

For $m = 4$ with $e_1 = 1$, the solution is $\beta_1 = 31$. If $e_1 = 2$, write $b = \beta_1$. Then $b > 4$, $b \equiv 2 \pmod{4}$, and the second increment N must be odd and positive. Equation (1) becomes $3(b - 1) + N = 90$, so $N = 93 - 3b > 0$ and $b \leq 30$. Thus b is exactly one of the seven stated values. The second characteristic exponent is $b + N = 93 - 2b$. These values satisfy the threshold inequality; no other b can occur.

For $m = 6$, the threshold inequality requires $\beta_1 \leq 66/7$, so the only integer first exponents greater than six are 7, 8, 9. With $\beta_1 = 7$ the gcd is one and $2\delta = 30 \neq 90$. With $\beta_1 = 8$, $e_1 = 2$, and (1) gives $N = 55$, hence $\beta_2 = 63$. With $\beta_1 = 9$, $e_1 = 3$, and $40 + 2N = 90$ gives $N = 25$, hence $\beta_2 = 34$. Both final gcds are one. This completes the exhaustive case split. $\square$

6.3 Eliminating the twelve entries

For each row below, S is the semigroup generated by the displayed positive integers. The indicated j violates (3); the columns give the actual and required counts. This is separate from the exhaustion proof.

Characteristic sequence	Generators of S	j	$R(11j + 1)$	Required
(2; 91)	2, 91	1	6	3
(3; 46)	3, 46	1	4	3
(4; 6, 81)	4, 6, 81	1	5	3
(4; 10, 73)	4, 10, 83	1	4	3
(4; 14, 65)	4, 14, 79	2	9	6
(4; 18, 57)	4, 18, 75	2	8	6
(4; 22, 49)	4, 22, 71	2	7	6
(4; 26, 41)	4, 26, 67	3	11	10
(4; 30, 33)	4, 30, 63	4	16	15
(4; 31)	4, 31	4	16	15
(6; 8, 63)	6, 8, 79	2	9	6
(6; 9, 34)	6, 9, 43	2	7	6

The small semigroup counts can be checked directly. For instance, below 45, $\langle 4, 31 \rangle$ contains the twelve multiples of 4 from 0 to 44, together with 31, 35, 39, 43, for a total of 16, rather than 15. The supplied program records the actual elements for every row and reproduces the counts by two different elementary enumeration routines.

Proof of Theorem 6.1. Rationality, the degree-genus formula, and uniqueness of the singular point give $\delta = g_{11} = 45$. If the threshold were at least $3/11$, Lemma 6.2 would place its characteristic sequence in the twelve-row list. Every row contradicts the necessary semigroup distribution identity. This is impossible. $\square$

As a cross-check with the literature rather than a dependency of this finite argument, the prior degree- ≤ 30 classification [5, Thm. 5.1, Table 7] leaves only the standard (10; 11) singularity in degree eleven. Its threshold $1/10 + 1/11 = 21/110$ is indeed below $3/11$.

Exact remaining gap. Theorem 6.1 concerns irreducible rational unicuspidal divisors on the ordinary plane. It says nothing about a reducible or nonreduced Calabi–Yau divisor, a curve on a singular Manetti surface, or multiple singular points. In particular it is not a classification of degree-11 smooth limits, and cannot justify marking the original problem solved. Extending the low-degree graph casework to degree eleven would require new proofs; the degree-5, 7 hypotheses in [2] are not silently extrapolated.

7 Conclusions and reproducibility

The five approaches give compatible necessary conditions and restricted positive cases, with no contradiction among them. They establish no new prime case of the original assertion. Our disposition is therefore: **unresolved after five substantive approaches**. Literature lookup, source inspection, code checks, and document preparation are not counted as mathematical approaches.

The accompanying original Python program uses only the standard library. It checks the limiting-net degree partitions; the hyperelliptic dimension formulas; the Markov and Cartier arithmetic; the nodal intersection obstruction; and the degree-11 cusp list, its exhaustion by a general gcd-chain enumeration, and every explicit semigroup witness. Two distinct semigroup algorithms agree. Negative controls reject a deleted candidate and an altered count. These exact finite checks support the stated proofs and are not a proof assistant certificate or independent peer review.

The immediate outstanding mathematical task is to control the full divisor and ambient surface in Hacking’s Calabi–Yau degeneration while preserving the smooth stable replacement. Any counterexample would have to be an actual smooth family with a special curve that admits no Cartier embedding on any allowed ambient surface; a numerical root, a low-gonality curve, or a singular plane equation alone would not suffice.

References

- [1] K. DeVleming (joint work with D. Stapleton), A question of Mori and families of plane curves, in *Algebraic Geometry: Moduli Spaces, Birational Geometry and Derived Aspects*, Oberwolfach Report 32/2022, contribution pp. 1821–1824; Conjecture B p. 1822. <https://doi.org/10.4171/owr/2022/32>.
- [2] K. DeVleming and D. Stapleton, *Smooth limits of plane curves of prime degree and Markov numbers*, J. Éc. polytech. Math. **11** (2024), 683–731. <https://doi.org/10.5802/jep.263>.
- [3] P. Hacking and Y. Prokhorov, *Smoothable del Pezzo surfaces with quotient singularities*, Compos. Math. **146** (2010), 169–192. <https://doi.org/10.1112/S0010437X09004370>; primary preprint <https://arxiv.org/abs/0808.1550>.
- [4] M. Borodzik and C. Livingston, *Heegaard Floer homology and rational cuspidal curves*, Forum Math. Sigma **2** (2014), e28. <https://doi.org/10.1017/fms.2014.28>; <https://arxiv.org/abs/1304.1062>.
- [5] K. DeVleming and N. Singh, *Rational unicuspidal plane curves of low degree*, arXiv:2311.15922v1 (2023), preprint. <https://arxiv.org/abs/2311.15922v1>.

- [6] K. DeVleming, J. Li, and S. Torres, *Weighted projective degenerations of $\mathbb{P}^n$* , arXiv:2606.21660v1 (2026), preprint. Relevant inspected application: §6.4. <https://arxiv.org/abs/2606.21660v1>.
- [7] K. DeVleming, *A guide to wall crossing for moduli of varieties*, arXiv:2602.20286v1 (2026), preprint. Relevant inspected application: §5.3. <https://arxiv.org/abs/2602.20286v1>.